

Create an AOT-Optimized Executable JAR of a Micronaut application

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CREATE AN AOT-OPTIMIZED EXECUTABLE JAR OF A MICRONAUT APPLICATION

Learn how to generate and run an executable JAR with Micronaut AOT optimizations using Maven or Gradle.

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Guide source content reviewed

common:header.adoc[]

common:requirements.adoc[]

== Create an Application

common:cli-or-launch.adoc[]

[source,bash]

```
----
mn @cli-command@ example.micronaut.micronautguide --build=@build@ --lang=@lang@ --features=micronaut-aot
----
```

common:build-lang-arguments.adoc[]

common:default-package.adoc[]

== Introduction

Micronaut AOT optimizes a Micronaut application at build time for a target runtime. For a JVM deployment, the build plugins can generate an optimized executable JAR that includes the application, its dependencies, and the AOT-generated classes and resources.

Use an AOT-optimized JAR when you want to keep a JVM deployment model but reduce runtime work such as environment deduction, property-source loading, and other startup-time analysis that can be performed during the build.

IMPORTANT: AOT produces an application optimized for the environment used during the AOT build. Build the optimized JAR with configuration and environment inputs that match the deployment environment.

external:distribution-base/hellocontroller.adoc[]

== Generate an AOT-Optimized Executable JAR

:exclude-for-build:maven

The https://micronaut-projects.github.io/micronaut-gradle-plugin/latest/#_micronaut_aot_plugin[Micronaut Gradle Plugin] adds optimized packaging tasks when the application includes Micronaut AOT support.

Generate an optimized executable JAR:

[source,bash]

```
----
./gradlew optimizedJitJarAll
----
```

The generated JAR is written to `build/libs/micronautguide-0.1-all-optimized.jar`.

:exclude-for-build:

:exclude-for-build:gradle

The <https://micronaut-projects.github.io/micronaut-maven-plugin/latest/examples/aot.html>[Micronaut Maven Plugin] enables AOT when you pass the `micronaut.aot.enabled` system property.

Generate an optimized executable JAR:

[source,bash]

```
----
./mvnw package -Dmicronaut.aot.enabled=true
----
```

The generated JAR is written to `target/micronautguide-0.1.jar`.

:exclude-for-build:

== Run the Optimized Executable JAR

Run the application packaged as an AOT-optimized JAR file:

:exclude-for-build:maven

[source,bash]

```
----
java -jar build/libs/micronautguide-0.1-all-optimized.jar
----
```

:exclude-for-build:

:exclude-for-build:gradle

```
[source,bash]
```

```
-----  
java -jar target/micronautguide-0.1.jar  
-----
```

:exclude-for-build:

You can verify the application responds by sending a request to the controller:

```
[source,bash]
```

```
-----  
curl localhost:8080  
-----
```

```
[source,json]
```

```
-----  
{ "message": "Hello World" }  
-----
```

== Time To First Request

=== Script

Use the following script to measure time to first request of a single output JAR.

resource:../../../../ttfr.sh[]

=== Measurement

Run the script against the optimized JAR:

```
[source,bash]
```

```
-----  
:exclude-for-build:maven  
./ttfr.sh -j build/libs/micronautguide-0.1-all-optimized.jar  
:exclude-for-build:  
:exclude-for-build:gradle  
./ttfr.sh -j target/micronautguide-0.1.jar  
:exclude-for-build:  
-----
```

Compare the result with a regular executable JAR from `guideLink:executable-jar[]` to decide whether the build-time optimization is worthwhile for your deployment.

common:next.adoc[]

Learn more about:

- * [https://micronaut-projects.github.io/micronaut-aot/latest/guide/\[Micronaut AOT\]](https://micronaut-projects.github.io/micronaut-aot/latest/guide/[Micronaut AOT])
- * https://micronaut-projects.github.io/micronaut-gradle-plugin/latest/#_micronaut_aot_plugin [Micronaut AOT support in the Micronaut Gradle Plugin]
- * <https://micronaut-projects.github.io/micronaut-maven-plugin/latest/examples/aot.html> [Micronaut AOT support in the Micronaut Maven Plugin]
- * `guideLink:executable-jar[]`

common:helpWithMicronaut.adoc[]